Siddhant Solanki

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EDUCATION

Stanford University *Ph.D. in Physics*	June, 2026 – Present
University of Maryland, College Park *Ph.D. in Astronomy*	June, 2024 – June, 2026 (Transfer)
University of Maryland, College Park *M.S. in Astronomy*	Aug, 2022 – June, 2024
University of California, Santa Barbara *B.S. in Physics*	Sept, 2017 – Dec, 2020

Research Interests

- Waves & Instabilities
- Wave–Particle Interactions
- Relativistic Plasmas
- Magnetars & Neutron Stars
- Fast Radio Bursts
- Black Hole Magnetospheres

SKILLS

- C, C++, Python
- OpenMP, MPI, Kokkos
- Paraview, Matplotlib
- Git
- High-Performance Computing
- Analytical Modeling of Complex Systems

PUBLICATIONS

Refereed Publications

1. **Solanki**, S., et al., Modeling of Light Curves from Reconnection-powered Very High Energy Flares from M87*, *ApJ*, 985, 147 (2025).
2. **Solanki**, S., et al., Short-period Heartbeat Binaries from TESS Full-frame Images, *ApJS*, 276, 17 (2025).
3. **Solanki**, S., et al., The Inner 2 pc of Sagittarius A*: Simulations of the Circumnuclear Disk and Multiphase Gas Accretion in the Galactic Center, *ApJ*, 953, 22 (2023).
4. **Solanki**, S., et al., Periodicities in the K2 Light Curve of HP Librae, *MNRAS*, 500, 1222 (2021).
5. Hakobyan, H., Böss, L., Cai, Y., Chernoglazov, A., . . . , Philippov, A., **Solanki**, S., Vanthieghem, A., Zhou, M., et al., *Entity*: Hardware-agnostic Particle-in-cell Code for Plasma Astrophysics. I. Curvilinear Special Relativistic Module, *ApJS*, 285, 11 (2026).

Preprints

1. **Solanki**, S., Mahlmann, J., Philippov, A., Nonlinear Decay of Fast Magnetosonic Waves through Weak Turbulence: Force-Free Electrodynamics Simulations, arXiv:2606.19450 (2026).
2. **Solanki**, S., Mahlmann, J., Philippov, A., Beloborodov, A., Damping of Fast Radio Bursts in the Inner Magnetospheres of Magnetars, arXiv:2606.19448 (2026).
3. Skrabacz, E., Murchikova, L., Ressler, S. M., Ukani, A., **Solanki**, S., Stellar-wind Fueled Accretion onto Sagittarius A*, arXiv:2601.09799 (2026).

AWARDS

1. Burgers Symposium Winner (2025)
   Awarded to the Best Poster in the College of Mathematical and Natural Sciences
2. John Chi-Lin Wang Award (2024)
   Awarded by the Astronomy Department at the University of Maryland
3. Academic High Honors Award (2021)
   Awarded by the Physics Department at UC Santa Barbara
4. Research High Honors Award (2021)
   Awarded by the Physics Department at UC Santa Barbara

LEADERSHIP AND OUTREACH

- DC Tutoring and Mentoring Initiative (2025-Present)
  Mentor in the Listo Program
- Graduate Student Council in the Astronomy Department (2023 - 2024)
- Cambridge Center for International Research (2023)
  Teaching Assistant
- Campus Learning and Assistance Services Tutor (2019-2021)

EXTERNAL CONFERENCES AND PRESENTATIONS

Invited Talks

- Kavli Institute for Theoretical Physics (KITP), UC Santa Barbara (2025)
  Invited Graduate Student Visitor
- Princeton Astroplasmas Seminar, Princeton University (2024)
  Invited Seminar Speaker
- Center for Computational Astrophysics, Flatiron Institute (2024)
  SPA Seminar Speaker
- Institute for Advanced Study (IAS), Princeton (2024)
  Coffee Talk Speaker

Summer Schools & Workshops

- SHIELD Plasma Summer School, Boston (2026)
  Participant
- PLASMA Science Summer School, Porquerolles, France (2024)
  Participant
- Les Houches Summer School in Plasma Physics, France (2023)
  Participant & Poster Presenter
- International High Performance Computing Summer School (2023)
  Participant & Poster Presenter

In the News

Monthly Roundup: The Supermassive Black Hole of Messier 87, AAS Nova (June 2025)
https://aasnova.org/2025/06/25/monthly-roundup-the-supermassive-black-hole-of-messier-87/